Haodong Yan

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Education

The Hong Kong University of Science and Technology (Guangzhou) — Guangzhou, China
Ph.D. in Intelligent Transportation — *Sep. 2024 – Present*

Xi'an Jiaotong University — Xi'an, China
Master of Mechanical Engineering — *Sep. 2021 – Jul. 2024*
Rank: 1/281
GPA: 3.73/4.0, Average Score: 89.68

Xi'an Jiaotong University — Xi'an, China
Bachelor of Mechanical Engineering — *Sep. 2017 – Jan. 2021*

GPA: 3.75/4.3, Average Score: 88.75

Xi'an Jiaotong University — Xi'an, China
Second Bachelor of Computer Science and Technology — *Jul. 2019 – Jul. 2021*

Selected Publications

[1] Yanjiang Guo*, **Haodong Yan***, Zhide Zhong*, Zhongru Zhang*, Qingyuan Yang*, Qingzhou Lu, Xiaoyu Chen, Yen-Jen Wang, Shuying Deng, Chenghan Yang, Puzhen Yuan, Chenxin Liu, Tun Ban, Xiang Zhu, Yichen Liu, Kun Feng, Haoang Li, Jianyu Chen, "Video Prediction Policy 2: Predict Better, Act Better," *arXiv*, 2026.
RoboDojo-Sim #1 overall (Oct. 8, 2026): 39.26 score, 32.26% success rate.
Project | Code | Models | Leaderboard

[2] **Haodong Yan**, Zhide Zhong, Jiaguan Zhu, Junjie He, Weilin Yuan, Wenxuan Song, Xin Gong, Yingjie Cai, Guanyi Zhao, Xu Yan, Bingbing Liu, Ying-Cong Chen, Haoang Li, "S-VAM: Shortcut Video-Action Model by Self-Distilling Geometric and Semantic Foresight," *ECCV*, 2026.

[3] **Haodong Yan***, Hang Yu*, Zhide Zhong*, Weilin Yuan, Xin Gong, Zehang Luo, Chengxi Heyu, Junfeng Li, Wenxuan Song, Shunbo Zhou, Haoang Li, "Open-world Hand-Object Interaction Video Generation Based on Structure and Contact-aware Representation," *CVPR*, 2026.

[4] Zhide Zhong*, **Haodong Yan***, Junfeng Li, Xiangchen Liu, Xin Gong, Tianran Zhang, Wenxuan Song, Jiayi Chen, Xinhu Zheng, Hesheng Wang, Haoang Li, "FlowVLA: Visual Chain of Thought-based Motion Reasoning for Vision-Language-Action Models," Preprint, under review at *IEEE Transactions on Robotics*.

[5] **Haodong Yan**, Pengxu Hou, Zhide Zhong, Xinhu Zheng, Zhe Liu, Hesheng Wang, Haoang Li, "CARE: Contextually-Aligned and Realistic 4D Scene Generation from a Single Image and Text," Preprint, under review at *IEEE Transactions on Image Processing*.

[6] **Haodong Yan**, Jiaguan Zhu, Mingyuan Jia, Ruiqing Yin, Junjie He, Zhide Zhong, Junfeng Li, Jinxuan Lu, Hengtao Li, Tianran Zhang, Jiayi Chen, Wenxuan Song, Wen Chen, Yuxiang Gao, Haoang Li, "PSG-JEPA: Is Forward Prediction Enough? Physical State Grounding for JEPA World Models," *arXiv*, 2026.

[7] **Haodong Yan**, Junfeng Li, Junjie He, Zhide Zhong, MingMing Yu, Wenxuan Song, Jiaguan Zhu, Yangyang Zheng, Yuqiao Du, Jiadi You, Yingjie Cai, Xu Yan, Guanyi Zhao, Bingbing Liu, Haoang Li, "Robust-WAM: Bridging Generative Pretraining and Semantic Foresight in World-Action Models," *arXiv*, 2026.

[8] Zhide Zhong*, **Haodong Yan***, Junfeng Li, Junjie He, Tianran Zhang, Haoang Li, "VLA-OPD: Bridging Offline SFT and Online RL for Vision-Language-Action Models via On-Policy Distillation," *arXiv*, 2026.

[9] Zhide Zhong, Junfeng Li, Junjie He, **Haodong Yan**, Xin Gong, Guanyi Zhao, Yingjie Cai, Jiantao Gao, Xu Yan, Bingbing Liu, Yingcong Chen, Liuqing Yang, Haoang Li, "DualCoT-VLA: Visual-Linguistic Chain of Thought via Parallel Reasoning for Vision-Language-Action Models," *arXiv*, 2026.

[10] Huashuo Lei, Wenxuan Song, Huarui Zhang, Jieyuan Pei, Jiayi Chen, **Haodong Yan**, Han Zhao, Pengxiang Ding, Zhipeng Zhang, Lida Huang, Donglin Wang, Yan Wang, Haoang Li, "RoboMemArena: A Comprehensive and Challenging Robotic Memory Benchmark," *arXiv*, 2026.

[11] Junfeng Li, Junjie He, Zhide Zhong, Yangyang Zheng, Pingyue Sheng, Jiayu Dong, Ruixin Li, **Haodong Yan**, Jiaguan Zhu, Tianran Zhang, Runze Yu, Wen Chen, Liuqing Yang, Yuxiang Gao, Haoang Li, "DyPES-VLA: Learning Shared Dynamics Priors and Embodiment-Specific Control for Cross-Embodiment Manipulation," *arXiv*, 2026.

[12] Junjie He, Junfeng Li, Zhide Zhong, **Haodong Yan**, Ruixin Li, Yangyang Zheng, Jiaguan Zhu, Tianran Zhang, Yuqiao Du, Wen Chen, Shunbo Zhou, Haoang Li, "SG-WAM: Text-Grounded and Spatial-aware Semantic Guidance for World-Action Models," *arXiv*, 2026.

[13] GigaWorld Team (incl. **Haodong Yan**), "GigaWorld-Policy-0.5: A Faster and Stronger WAM Empowered by AutoResearch," *arXiv*, 2026.

[14] Wenxuan Song, Jiayi Chen, Xiaoquan Sun, Huashuo Lei, Yikai Qin, Wei Zhao, Pengxiang Ding, Han Zhao, Tongxin Wang, Pengxu Hou, Zhide Zhong, **Haodong Yan**, Donglin Wang, Jun Ma, Haoang Li, "Rethinking the Practicality of Vision-language-action Model: A Comprehensive Benchmark and An Improved Baseline," *ICRA*, 2026.

[15] Wenxuan Song, Ziyang Zhou, Han Zhao, Jiayi Chen, Pengxiang Ding, **Haodong Yan**, Yuxin Huang, Feilong Tang, Donglin Wang, Haoang Li, "ReconVLA: Reconstructive Vision-Language-Action Model as Effective Robot Perceiver," *AAAI*, 2026 (**Outstanding Paper Award**).

[16] **Haodong Yan**[*], Zhiming Hu[*], Syn Schmitt, Andreas Bulling, "GazeMoDiff: Gaze-guided Diffusion Model for Stochastic Human Motion Prediction," *Pacific Graphics*, 2024.

[17] Cheng Lei, Junpeng Hu, **Haodong Yan**, Mariia Gladkova, Tianyu Huang, Yun-Hui Liu, Daniel Cremers, Haoang Li, "Physically-Based Photometric Bundle Adjustment in Non-Lambertian Environments," *IROS*, 2024.

[18] **Haodong Yan**, Fudong Li, Jinglong Chen, Zijun Liu, Jun Wang, Yong Feng, Xinwei Zhang, "A Graph Embedded in Graph Framework with Dual-sequence Input for Efficient Anomaly Detection of Complex Equipment under Insufficient Samples," *Reliability Engineering & System Safety*, 2023.

[19] **Haodong Yan**, Zijun Liu, Jinglong Chen, Yong Feng, Jun Wang, "Memory-augmented Skip-connected Autoencoder for Unsupervised Anomaly Detection of Rocket Engines with Multi-source Fusion," *ISA Transactions*, 2023.

[20] **Haodong Yan**, Jun Wang, Jinglong Chen, Zijun Liu, Yong Feng, "Virtual Sensor-based Imputed Graph Attention Network for Anomaly Detection of Equipment with Incomplete Data," *Journal of Manufacturing Systems*, 2022.

[*] equal contribution.

Work Experience

RobotEra — Beijing, China
Internship — *Jul. 2026 – Present*

- Vision-Language-Action and World Models for robot manipulation

GigaAI — Beijing, China
Internship — *May 2026 – Jul. 2026*

- World-Action Model (WAM) foundation model training (GigaWorldPolicy)

Institute for Visualisation and Interactive Systems, University of Stuttgart — Stuttgart, Germany
Internship — *Jul. 2023 – Oct. 2023*

- Code implementation
- Paper writing

Computer Vision Group, Technical University of Munich — Remote
Remote Internship — *Apr. 2023 – Dec. 2023*

- Code implementation
- Paper writing
- Data processing

Honours and Awards

Outstanding Paper Award – AAAI (ReconVLA) 2026

National Scholarship – Ministry of Education, China 2023

National Scholarship – Ministry of Education, China 2022

China Optics Valley Scholarship – Wuhan High-tech Zone Government 2022

Outstanding Graduates – Xi'an Jiaotong University 2021

HIWIN Elite Scholarship – HIWIN Technologies Corp. 2021

National First Prize – Chinese University Students Mechanical Innovation Competition 2020

National First Prize – Chinese Intelligent Robot Fighting Competition 2019

Skills

Programming: Python (PyTorch, OpenCV, Transformers), C++, LaTeX, Git.

Research Interests: Robot Learning (Video-Action Models, World Models), Video Generation Models.

Languages

English: IELTS Overall 7.0 (Listening 7.0, Reading 8.5, Writing 6.5, Speaking 6.0).

Mandarin Chinese: Native.